

LESSON 7-7 • SESSION 1 • PRACTICE • FORM B

Polygons and Perimeter on the Plane

Learning target – I can find the perimeter and area of a polygon drawn on the coordinate plane. · 6.G.A.3

REMEMBER

- A **polygon** is a closed shape with straight sides. Its corners are **vertices**.
- Find each side from matching coordinates: opposite signs **add**, same sign **subtract**.
- **Perimeter** = add all the sides. **Area** of a rectangle = length $\times$ width.

WORKED EXAMPLE

A rectangle has vertices $(-2, 3)$, $(5, 3)$, $(5, -2)$ and $(-2, -2)$. Each unit is 1 meter. Find its perimeter.

Step 1 Top side: same y , so $2 + 5 = 7$ m.

Step 2 Right side: same x , so $3 + 2 = 5$ m.

Step 3 Add all four sides: $7 + 5 + 7 + 5 = 24$.

Answer Perimeter = 24 m

WORDS polygon (*polígono*) · vertex (*vértice*) · perimeter (*perímetro*)

PRACTICE

show your work · label units

1. A rectangle has vertices $(-5, 3)$, $(5, 3)$, $(5, -4)$ and $(-5, -4)$. Find each measure.

a. side $(-5, 3)$ to $(5, 3)$ _____

b. side $(5, 3)$ to $(5, -4)$ _____

c. perimeter _____

d. area _____

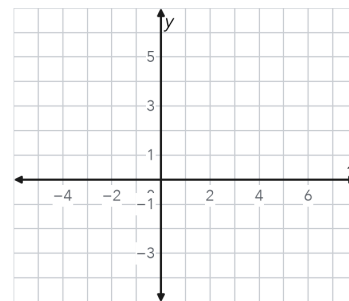
2. **Garden Plot.** A garden plot has corners at $(-5, 5)$, $(7, 5)$, $(7, -4)$ and $(-5, -4)$. Each unit is 1 meter.

a. Plot the corners and connect them.

b. Top side _____ m

c. Left side _____ m

d. Perimeter _____ m



3. **Dog Run.** Dog run vertices: $(-6, 9)$, $(10, 9)$, $(10, -3)$ and $(-6, -3)$. Each unit is 1 meter. Find the perimeter.

Hint Find one side across and one side up and down.

Show your work.

Answer _____ m

4. **Missing Vertex.** Three corners of a rectangle are $(-3, 8)$, $(7, 8)$ and $(7, 2)$. Name the fourth corner.

Hint Match x with one corner and y with another.

Show your work.

Corner _____

5. **Right Triangle.** A right triangle has vertices $(-4, 3)$, $(6, 3)$ and $(6, -5)$. Find its area.

Hint Area = $\frac{1}{2} \times \text{base} \times \text{height}$.

Show your work.

Answer _____ sq. units

6. **Deck Area.** Deck vertices: $(-5, -1)$, $(8, -1)$, $(8, -7)$ and $(-5, -7)$. Each unit is 1 yard. Find its area.

Hint Both y -values are negative, so subtract to find the height.

Show your work.

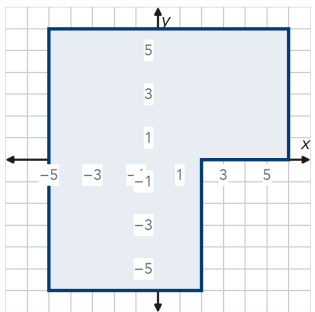
Answer _____ sq. yd

7. **Mural Outline.** The mural outline has vertices $(-5, 6)$, $(6, 6)$, $(6, 0)$, $(2, 0)$, $(2, -6)$ and $(-5, -6)$. Each unit is 1 foot. Find the perimeter.

Hint Go around in order. Each side joins two vertices that share a coordinate.

Show your work.

Answer _____ ft



8. **Fence and Sod.** A plot has vertices $(-5, 6)$, $(9, 6)$, $(9, -3)$ and $(-5, -3)$. Fencing costs \$6 per unit. Sod costs \$3 per square unit.

- Find the length and the width. _____
- Find the perimeter and the cost of fencing. _____
- Find the area and the cost of sod. _____

FIND THE ERROR

explain the mistake, then fix it

9. **Bri's Work.** For a side from $(-9, 2)$ to $(6, 2)$, Bri wrote the length as $9 - 6 = 3$ units. Explain her mistake. Then find the correct length.

TEST PRACTICE

choose the best answer

10. A rectangle with sides 9 units and 4 units has a perimeter of –
A 13 units **B** 26 units **C** 36 units **D** 18 units
11. A rectangle with sides 9 units and 4 units has an area of –
A 13 square units **B** 26 square units **C** 36 square units **D** 72 square units

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

40 R	46 E	80 Z	78 N	\$138 J	56 C	104 X	$(-3, 2)$ O	\$276 R	28 Q
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#3	#4	#5	#6	#7	#8b

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____